MECHANISM DEEP DIVE · NEW DIAGNOSTICS

Why steering collapses and screening survives

The methods deck said *what* each component is. This one traces the causal chain from "B solver calls" to "which method wins", and tests every link with a measurement on the actual trained model.

Every figure here is new: none appears in the paper.

How this deck differs from the methods deck

The paper asserts a mechanism — *diversity collapse* — and supports it indirectly, by noting that steering fails to improve as B grows. That is an inference from an outcome, not a measurement of a cause.

Here every link in the chain is instrumented on the trained regime-B model:

claim in the chain	how it is tested here
extra budget only helps via distinct basins	basin coverage vs. B, four methods
steering makes seeds near-identical	PCA of 60 optimization paths; pairwise distances
the trust region causes the collapse	sweep τ over seven values
the surrogate is honest only near the prior	calibration vs. $\|z\|$
screening needs only ranking	rank correlation vs. actual selection quality

Two of these measurements revise what the paper says. Those slides are marked.

PART I

What a Budget Actually Buys

The unit of value is not a seed

A strategy produces B seeds $x_0^{(1..B)}$, solves all of them, and returns

$$c \;=\; \min_{j=1..B} \; \mathrm{solve}\big(x_0^{(j)};\theta\big).$$

The solver is a many-to-one map

Local descent sends every seed in a basin of attraction to that basin's minimum. So the B solved costs take at most as many *distinct values as there are distinct basins* among the B seeds.

Two seeds in the same basin return the same number. The min over B of them is that number.

Therefore: budget is only converted into value through basin diversity. A method that produces 16 seeds in one basin has spent 16 queries to buy exactly what 1 query buys. This is the single fact the rest of the deck hangs on.

The same point as order statistics

Let p be the probability that one seed lands in a basin at least as good as some target.

Independent seeds

$$\Pr[\text{all } B \text{ miss}] = (1-p)^B$$

Failure decays *geometrically* in B. Doubling the budget squares the miss probability.

Perfectly correlated seeds

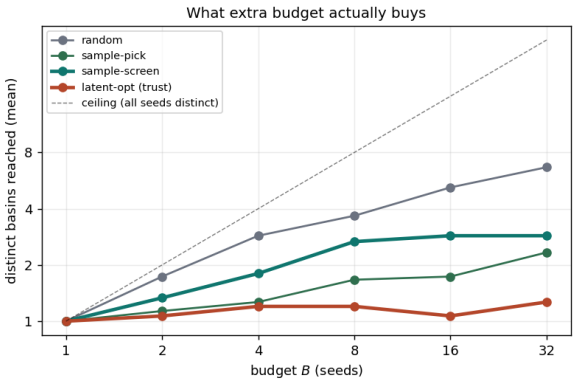
$$\Pr[\text{all } B \text{ miss}] = (1-p)$$

Failure is *constant* in B. Budget buys nothing at all.

Real methods live between these two extremes, and where they live is a property of *how the seeds were generated*, not of the model's quality.

So the budget-scaling curve of a method is a direct readout of the statistical dependence among its seeds. A flat curve is not a sign of a weak model — it is a sign of **dependent seeds**. That is what we should expect steering to show, and it is what we will measure next.

Measured: what extra budget buys each method [measured here]



Mean number of distinct basins reached by B seeds, regime B, shifted instances, 15 instances per point. Basin identity = nearest well center to the solved solution.

	B=1	B=32
random	1.00	**6.67**
sample-screen	1.00	2.87
sample-pick	1.00	2.33
latent-opt (τ=1)	1.00	**1.27**

Steering's curve is **flat**: 32 seeds reach 1.27 basins. It is spending 32 queries to buy roughly one.

The mechanism the paper asserts is real, and this is the direct evidence for it.

But coverage alone is not the resource [measured here]

Look again at the previous slide: `random` **has by far the most basin coverage** — 6.67 distinct basins at B=32, more than double screening's — and `random` is the worst-performing method in the paper.

So "diversity" cannot be the quantity that matters

Random seeds are maximally spread and land in many basins — *arbitrary* ones, mostly shallow. Coverage of bad basins is worthless.

method	coverage	seed quality	outcome
random	high	low	many shallow basins
sample-pick	low–mid	mid	few basins, decent ones
latent-opt (trust)	very low	high	one good basin, B times over
sample-screen	mid	high	**several good basins**

The resource is the product of the two: distinct *plausible* basins. Screening is the only method that raises one without destroying the other — and that is the whole story in one line.

A refinement of the paper's claim [measured here]

The paper writes that steering "forfeits the diversity that scarce-budget multi-start depends on". The measurement says that is right in spirit and imprecise in wording.

As written

Diversity is the resource; steering destroys it.

Under that reading, `random` — the most diverse method by a wide margin — should be the strongest. It is the weakest.

As measured

Diverse plausible seeds are the resource. Two ingredients, and a method needs both.

This explains all four methods' behaviour with one rule instead of separate excuses for `random`.

The distinction matters for the paper's own recommendation. "Add a repulsion term to steering" only works if the repulsion pushes seeds toward *other good* basins. Repulsion that merely spreads seeds out reproduces `random`, which we now know is the worst thing to be.

PART II

Why Steering Collapses

The mechanism, stated precisely

All B latents are optimized against **the same objective**, for the same instance:

$$z^{(j)} \leftarrow \arg\min_{z} \; g(z, \theta) + \tau \|z\|^2, \qquad j = 1, \ldots, B,$$

starting from independent draws $z_0^{(j)} \sim \mathcal{N}(0, I)$ — and **nothing couples them**. No repulsion, no shared state, no diversity term.

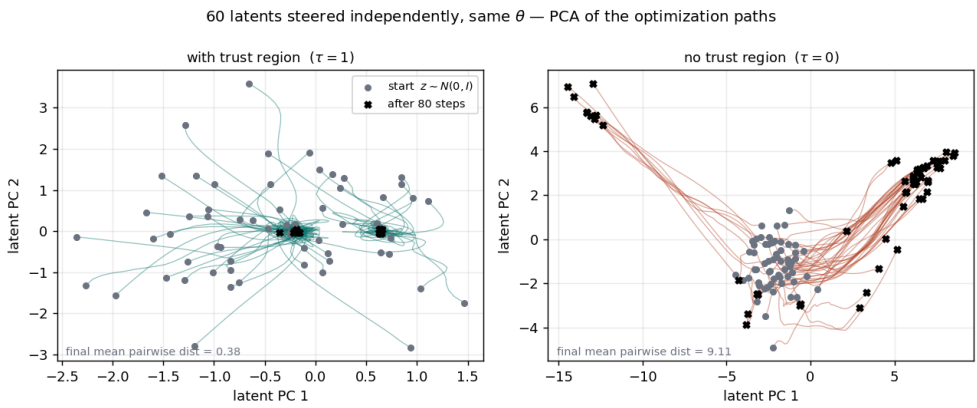
What gradient descent does to a set of points

Adam on a fixed smooth objective is (approximately) a deterministic flow. Such a flow partitions the space into basins of attraction of $g + \tau\|\cdot\|^2$. Every start inside one basin converges to the same attractor.

$$\#\{\text{distinct } z^\star\} \;\leq\; \#\{\text{attractors of the surrogate reached}\} \;\ll\; B.$$

So the collapse is not a bug in the implementation and not a symptom of a weak surrogate. It is what running B copies of the same optimizer *is*. And $\tau\|z\|^2$ makes it worse: it adds a single attractor at the origin that every trajectory feels.

Measured: 60 latents, one instance, 80 steps [measured here]



Left (τ=1): 60 independent starts (grey) funnel into *two* tight clusters. Final mean pairwise latent distance **0.38**. Sixteen of these would be sixteen copies of about two points. **Right** (τ=0): the same optimizer without the penalty flies outward to $\|z\| \sim 14$ — final spread **9.11**, and every one of those points is off-manifold, where the surrogate's promises are worthless.

Two failure modes, one dial

The trajectory figure shows something the paper's Figure 2 alone does not: **the trust region is what causes the collapse**. Turning τ down restores diversity — and destroys validity at the same time.

τ small: diverse but invalid

Latents scatter far from the prior. g was never trained there, promises cost it cannot deliver, and the decoded seeds are meaningless. Diversity of garbage.

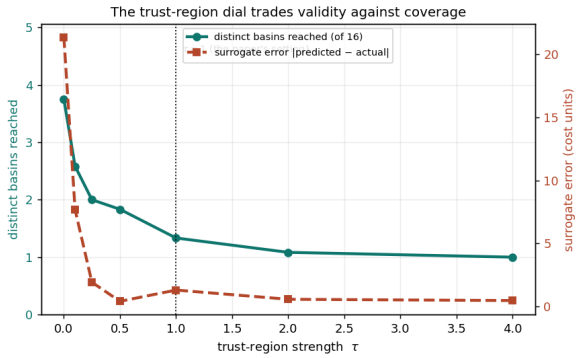
τ large: valid but identical

Latents are pulled onto the prior where g is honest — and onto *each other*. Validity without coverage.

Both failure modes are the same phenomenon seen from two sides: the penalty is a *shared* attractor, so it buys honesty by making the seeds agree.

If that is right, then τ should trade the two off monotonically — and no setting should deliver both. That is a testable prediction, so let us test it.

Measured: sweeping the trust region [measured here]



B=16, regime B, shifted, 12 instances.

τ	seed spread	basins	|pred−actual|
0.00	13.41	3.75	21.37
0.10	4.78	2.58	7.68
0.25	1.82	2.00	1.93
0.50	0.76	**1.83**	**0.41**
1.00	0.35	1.33	1.31
2.00	0.05	1.08	0.58
4.00	0.02	1.00	0.47

Coverage falls monotonically with τ (3.75 $\rightarrow$ 1.00) while surrogate error collapses (21.4 $\rightarrow$ 0.5). The prediction holds: **one dial, two goods, no setting gives both**.

An uncomfortable detail the paper should address [measured here]

Read the sweep table again at two rows:

τ	distinct basins	surrogate error	
0.50	**1.83**	**0.41**	better on *both* axes
1.00 (the paper's setting)	1.33	1.31	

$\tau = 0.5$ dominates $\tau = 1$: more coverage *and* more honest predictions. The paper fixes $\tau = 1$ with no sweep reported, so part of steering's measured weakness is a hyperparameter choice rather than a structural fact.

What this does and does not overturn

It does *not* rescue steering: even the best row (1.83 basins) sits below screening's 2.87 with comparable seed quality, and the monotone trade-off means no τ escapes. But the honest statement is "steering loses across a swept τ", and the paper is currently only entitled to "steering loses at $\tau = 1$".

Recommendation: run the sweep, report it, and quote the best τ. The conclusion survives; the claim gets stronger.

Why more steering effort makes it worse

A natural objection: perhaps steering is simply under-optimized — give it more than 80 steps.

The mechanism predicts the opposite. Under a descent flow, additional steps move each trajectory *further along* toward its attractor. More steps means:

- tighter clustering at each attractor (less coverage);
- and, at low τ, further travel from the prior (less validity).

The awkward asymmetry

For screening, spending more free compute — a larger pool — monotonically *helps*: more candidates to rank, same diversity, better tail. For steering, spending more free compute monotonically *hurts* on the axis that matters.

This is why the paper's budget-scaling observation is diagnostic rather than incidental. A method whose failure mode intensifies with effort cannot be tuned out of it; it has to be *restructured* — by coupling the B optimizations to each other.

PART III

Why Screening Survives

Screening is a filter, not a map

Steering transports points.

$z \mapsto z^\star$. The output distribution is whatever the flow produces — concentrated, and possibly far from anywhere the model was trained.

Screening conditions a sample.

The 64 candidates are i.i.d. prior draws. Selection keeps a subset. No candidate is ever *moved*.

Three consequences follow immediately, and they are the reasons screening is robust:

1. **Every surviving seed is a genuine model sample.** Nothing is off-manifold, because nothing left the manifold. The failure mode of Figure 2 cannot occur.
2. **Spread is preserved by construction.** Selection cannot make two distinct candidates more similar.
3. **The surrogate is only ever asked to compare points inside its training support.** It is an interpolator, not an extrapolator.

Screening even *raises* coverage relative to `sample-pick` (2.87 vs 1.73 at $B{=}16$): poor decodes disproportionately fall into the same default basin, so filtering them out leaves a more varied set.

What does screening actually demand of the surrogate?

Steering consumes g's **gradient** and its **values**, off-support. Screening consumes only an **ordering**, on-support — and not even a full ordering.

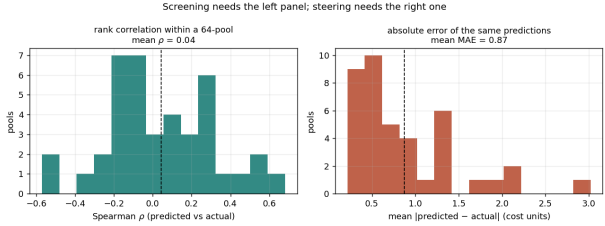
Precisely: `sample-screen` succeeds if and only if the good candidates appear in the top B. It does not matter

- whether predicted costs are numerically accurate,
- whether the middle of the pool is ordered correctly,
- or whether the scale is right — any monotone transform of g gives identical behaviour.

Which suggests the obvious test — and a trap

The obvious way to check a ranker is a rank correlation over the pool. That is the wrong statistic here, and it is wrong in a way that would make screening look broken.

Measured: the wrong statistic [measured here]



Over 40 pools of 64 candidates, shifted instances:

- mean Spearman $\rho = +0.047$ — statistically indistinguishable from no ordering ability at all;
- mean absolute error ≈ 0.9 cost units, on gaps of order 3–7.

Taken at face value, these two numbers say the ranker is useless under distribution shift — which would make screening's success inexplicable. The numbers are correct. The interpretation would be wrong.

Why $\rho \approx 0$ and a good screener are consistent

Inside a 64-candidate pool there are only **2.9 distinct basins** on average, and just **14%** of candidates sit in the best one. So the pool's actual costs are nearly a step function: a few good values, a large block of near-ties.

Where a global rank correlation spends its attention

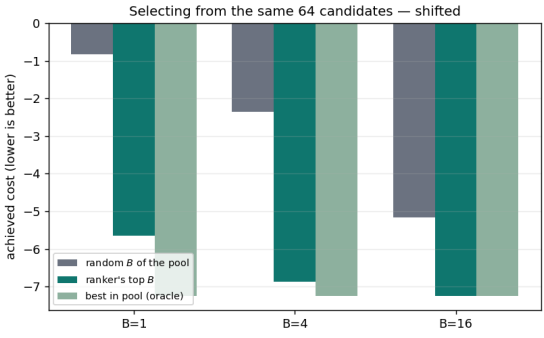
Spearman is dominated by the *majority* of pairs — and most pairs are two near-tied mediocre candidates whose relative order is pure noise. It measures ordering ability in exactly the region nobody cares about, and is almost blind to the top tail, which is the only region screening uses.

The right question is not "does g order the pool?" but:

does the top-B set that g selects contain the good candidates?

That is a statement about the tail, and it must be measured against the correct baseline: a random B-subset of the *same* pool.

Measured: the right statistic [measured here]



Same 40 shifted pools. Cost achieved by three selection rules over identical candidates:

	B=1	B=4	B=16
random B	−0.83	−2.36	−5.16
ranker top-B	−5.64	−6.88	−7.26
oracle (pool best)	−7.26	−7.26	−7.26

And the probability that the selected set contains the pool's single best candidate:

	B=1	B=4	B=16
ranker	**0.35**	**0.70**	**0.90**
chance	0.016	0.06	0.26

At B=1 the ranker finds the best of 64 about **22× more often than chance**; at B=16 it matches the oracle exactly.

The robustness property, isolated

Compare the same two statistics in and out of distribution:

	in-distribution	shifted
global rank correlation ρ	$+0.52$	$+0.05$
P(top-1 is the pool's best), chance $= 0.016$	0.28	**0.35**
cost achieved at $B{=}1$ vs random-1	-2.98 vs -1.13	-5.64 vs -0.83

Under shift the surrogate's **global ordering ability collapses** — and its **tail-identification ability does not**. Screening only consumes the second.

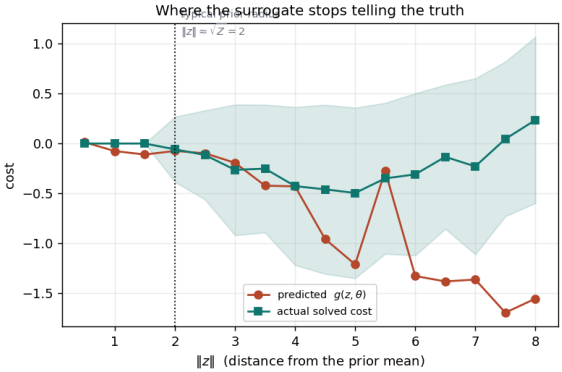
This is the real reason screening survives distribution shift

Not that the surrogate stays accurate — it does not. But shift degrades the fine-grained ordering among mediocre candidates long before it degrades the coarse "this one is in a deep well" signal. Screening asks only the coarse question, and the coarse question is the robust one.

PART IV

The Surrogate's Two Jobs

Measured: where the surrogate stops telling the truth [measured here]



64 latents at each fixed radius $\|z\| = r$, decoded and solved.

	predicted	actual
$r = 2$ (typical prior)	-0.07	-0.06
$r = 8$	-1.56	$+0.24$

At the prior's typical radius $\sqrt{Z} = 2$, prediction and reality agree to two decimals. By $r = 8$ the surrogate promises a cost it misses by 1.8 — in the optimistic direction, always. **Why:** g was trained on $z \sim \mathcal{N}(0, I)$. Beyond a couple of standard deviations it has no data, and an unconstrained MLP extrapolates confidently and arbitrarily.

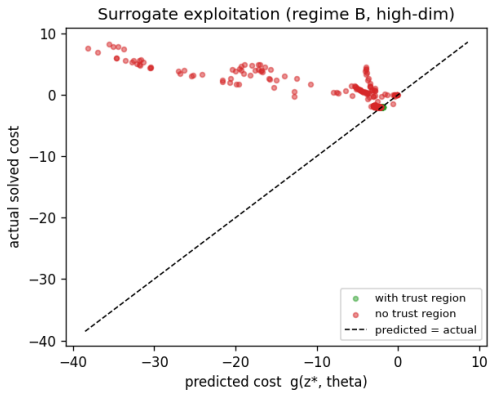
Random directions understate the danger

The calibration slide sampled *random* directions at each radius, and found a 1.8-unit over-promise at $r = 8$.

Gradient descent does not sample random directions. It **searches for the direction of steepest predicted improvement** — which is precisely the direction in which the surrogate is most wrong, because that is what "low predicted cost with no data" looks like.

The paper's Figure 2 shows the result of that search: predicted costs near -38 against actual costs around $+5$.

An optimizer pointed at a learned objective is an adversary against that objective's errors. The trust region is what makes the arrangement viable at all.



Paper Figure 2. Red: $\tau = 0$. Green: $\tau = 1$. Points below the diagonal are promises the surrogate cannot keep.

The two jobs, side by side

	steering needs	**screening needs**
what it consumes	values *and* gradients	an ordering
where	wherever the flow goes — systematically off-support	only at prior samples — always on-support
resolution	fine: small differences in g steer the trajectory	coarse: only the top tail must be right
failure under shift	compounds: extrapolation on top of shift	degrades gracefully: coarse signal survives
invariance	none: rescaling g changes the path	any monotone transform of g is identical

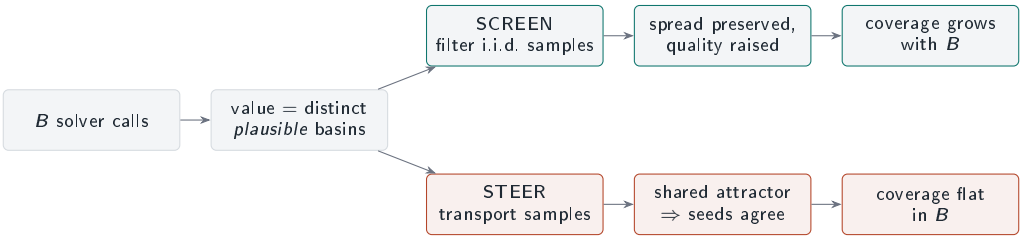
Same network, same weights, same training data. **The two arms differ entirely in how much they ask of it**, and the study's outcome follows from that asymmetry rather than from anything about latent geometry.

This is also why the paper's §5.4 result — swap the VAE for a diffusion model and nothing changes — is exactly what the mechanism predicts: the generator was never the active ingredient.

PART V

Synthesis

The whole causal chain



Every arrow in this chain is measured, not assumed: coverage curves (link 3), trajectory PCA and the τ sweep (link 2, lower path), selection quality and calibration (link 2, upper path). Nothing in the chain mentions the latent's geometry or the choice of generator — which is why changing the generator changes nothing.

Predictions the mechanism makes

A mechanism is only worth having if it forecasts things not yet measured. This one does:

1. **Screening's advantage should track pool size, not model quality.** Shrink the pool from 64 toward B and screening must degrade continuously into `sample-pick`.
2. **A repulsion term should move coverage and gap together.** If diversified steering raises coverage at $B{=}16$ from 1.33 toward 2.87 without losing seed quality, its gap should approach screening's. If coverage rises and the gap does not follow, the mechanism is wrong.
3. **Screening should degrade with *tail* ranking accuracy, not with ρ or MAE.** Corrupting g in ways that preserve the top tail should barely hurt it.
4. **Steering should worsen monotonically with steps.** 40, 80, 160, 320 surrogate steps $\Rightarrow$ monotonically falling coverage.
5. **Raising K (more wells) should widen the gap between screening and steering**, since more basins means more for coverage to buy.

Items 2 and 3 are the paper's own proposed follow-ups — now with a quantitative target and a falsification condition attached.

If you wanted to fix steering

The mechanism says exactly what is broken: B **independent copies of one deterministic optimizer**. So the fix must couple them.

Plausible couplings

- pairwise repulsion in latent space, $-\gamma \sum_{i<j} \|z_i - z_j\|$
- a determinantal / volume term over the batch
- quality-diversity: maintain an archive keyed by predicted basin
- simplest of all: steer *one* seed and let the other $B - 1$ be screened samples

What to measure

- coverage vs. B — the primary metric, not the gap
- seed quality alongside it, so you can tell repulsion from randomization
- the τ sweep, since coupling changes the trade-off

The trap: repulsion that merely spreads seeds reproduces `random`, which has the best coverage and the worst results.

Note the fourth bullet on the left. A hybrid is the honest baseline for any diversified-steering paper — and if steering cannot beat "one steered seed plus screened samples", it has no case.

What these diagnostics do not establish

- **One world, one training seed.** Every figure comes from the same regime-B world built with seed=0. The trends are large and consistent, but no error bars over training randomness are reported — the same gap the paper has.
- **Basin identity is a proxy.** A solved point is assigned to its nearest well center. With K=12 wells and a mild bowl this is usually right, but it is an approximation, and it would need rethinking on a landscape without explicit centers.
- **Small instance counts.** Coverage curves use 15 instances, the τ sweep 12, selection statistics 40 pools. Adequate for effects this size; not for fine distinctions.
- **The τ sweep is at one budget and one shift setting** (B=16, shifted). The monotone trend is clear, but the exact location of the knee is not pinned down.
- **Correlational, not causal, in one place**: we show coverage tracks method, and that the paper's gaps track coverage — but the clean causal test is the repulsion experiment in prediction 2, which has not been run.

The mechanism in one slide

1. Budget converts to value only through distinct plausible basins

The solver is many-to-one: B seeds in one basin are worth one seed. So budget scaling is a readout of statistical dependence among seeds — and coverage of *bad* basins is worthless, which is why `random` leads on diversity and loses on results.

2. Steering makes the seeds dependent, by construction

B copies of one deterministic optimizer share attractors; the trust penalty adds one more at the origin. Measured: coverage flat at ≈ 1.3 from $B{=}1$ to $B{=}32$; 60 latents collapse to two points. The dial that restores diversity ($\tau \to 0$) destroys validity, monotonically.

3. Screening keeps them independent, and asks less of the surrogate

Filtering cannot move a point or shrink a spread, and it needs only the coarse top-tail ordering — which survives distribution shift even when global rank correlation collapses to $\rho = 0.05$.

None of this refers to latent geometry. That is why swapping the generator changed nothing.